**Aufgabe 1:** (20 Punkte)

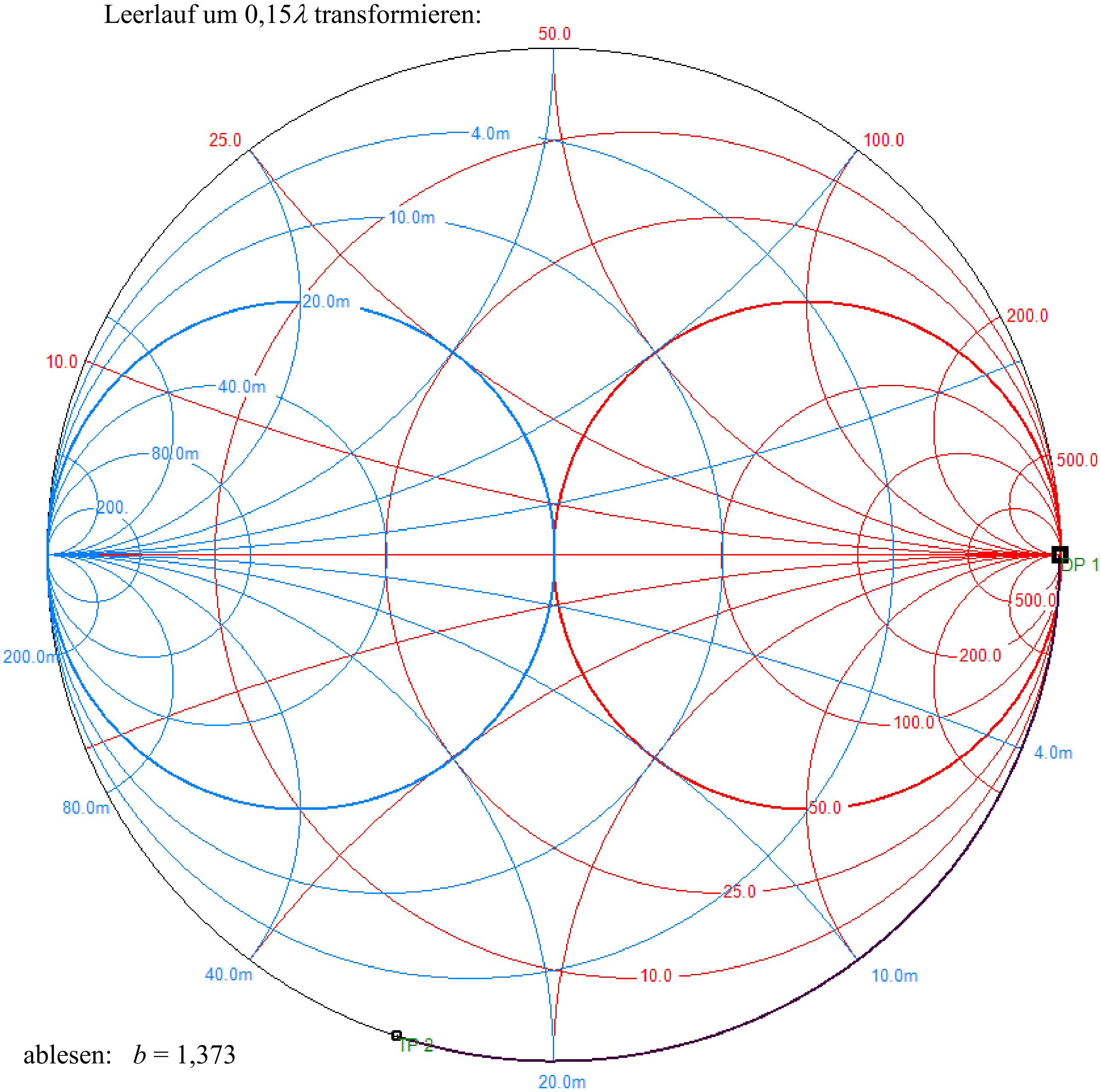
1.
$$r_A = \frac{R_A}{Z_L} = \frac{60\Omega}{50\Omega} = 1{,}2 \qquad x_{L_A} = \frac{j\cdot 2\pi\cdot f\cdot L_A}{Z_L} = \frac{j\cdot 2\pi\cdot 183\cdot 10^6\,\frac{1}{\mathrm{s}}\cdot 25\cdot 10^{-9}\,\frac{\mathrm{Vs}}{\mathrm{A}}}{50\Omega} = \frac{j28{,}75\Omega}{50\Omega} = j0{,}575$$

$$\underline{z}_A = 1{,}2 + j0{,}575 = 1{,}33\cdot e^{j25{,}6^\circ} \qquad \left(\text{Lastreflexionsfaktor } \underline{r}_A = 0{,}268\cdot e^{j56{,}2^\circ}\right)$$

$$\lambda = \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \frac{c_0}{f\cdot\sqrt{\varepsilon_r}} = \frac{3\cdot 10^8\,\frac{\mathrm{m}}{\mathrm{s}}}{183\cdot 10^6\,\frac{1}{\mathrm{s}}\cdot\sqrt{4{,}2}} = 800\mathrm{mm}$$

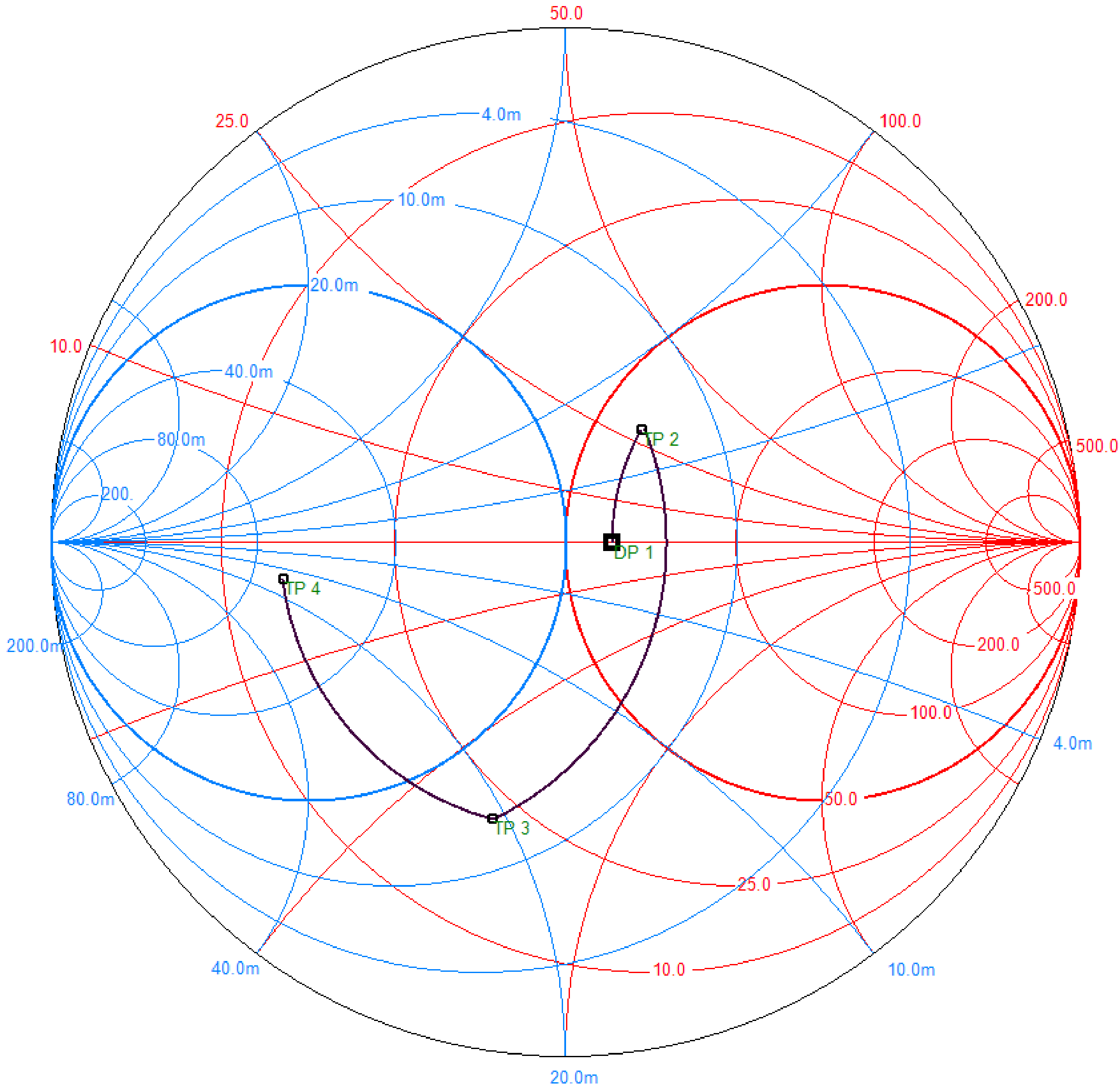
$$\frac{l_1}{\lambda} = \frac{120\mathrm{mm}}{800\mathrm{mm}} = 0{,}15 \qquad \frac{l_2}{\lambda} = \frac{475\mathrm{mm}}{800\mathrm{mm}} = 0{,}5938 \qquad \lambda/2 \text{ abgezogen: } 0{,}0938$$

Leerlauf um $0{,}15\lambda$ transformieren:



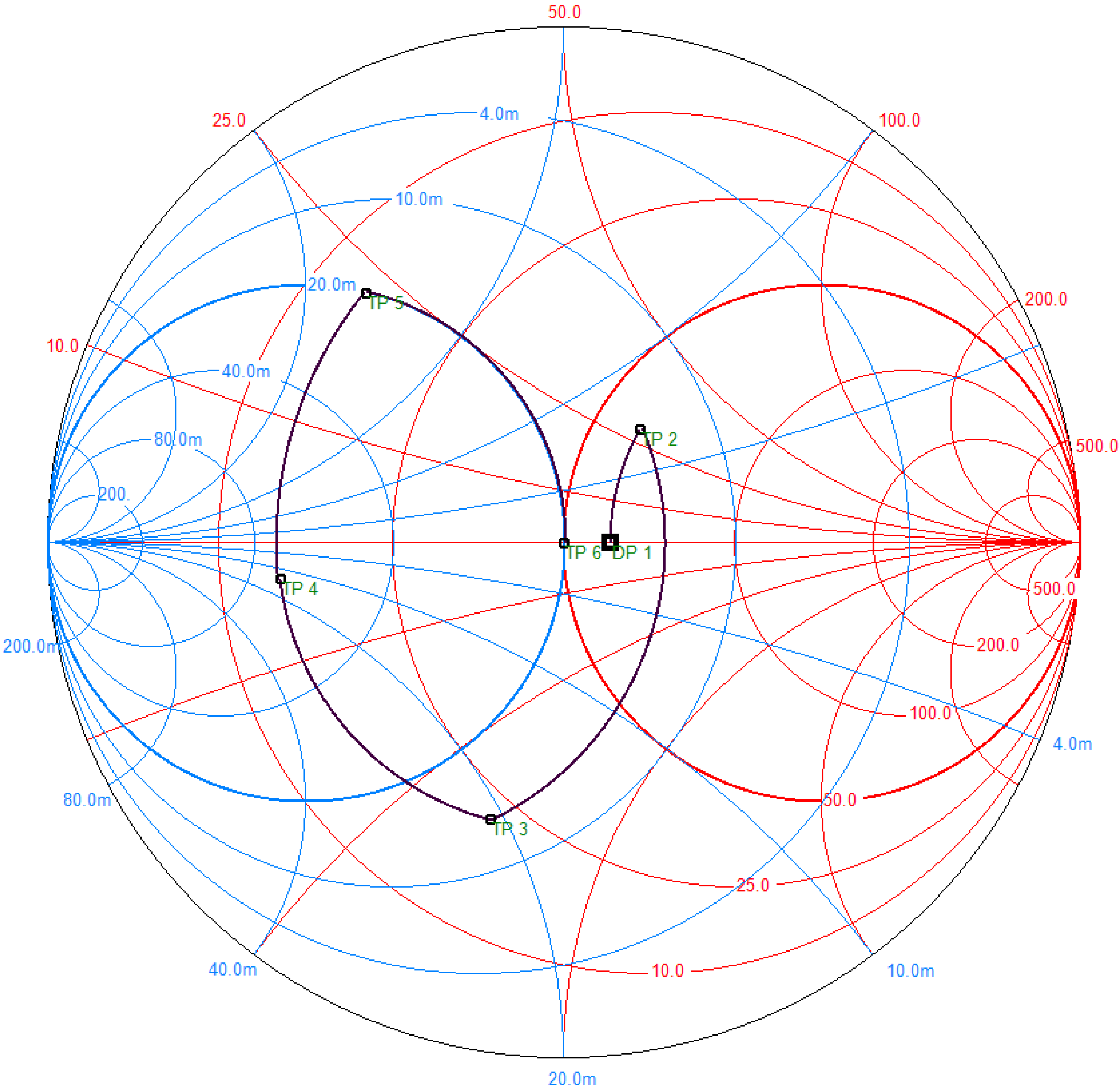

ablesen: $b = 1{,}373$

$\underline{z}_A$ einzeichnen in Smith-Chart, $b$ parallelschalten (TP3: $l/\lambda = 0{,}3958$) und um $l_2$ transformieren (bis $l/\lambda = 0{,}3958 + 0{,}0938 = 0{,}4906$)




ablesen: $|\underline{r}| = 0{,}556$ $\varphi = -172{,}7°$ $\underline{\underline{\underline{r} = 0{,}556 \cdot e^{-j172{,}7°}}}$ $[= -0{,}551 - j0{,}0755]$

2. Transformation über Serien-Induktivität $L$ und Parallel-Kapazität $C$:




$$X_L = \left[0{,}452 - \left(-0{,}0582\right)\right] \cdot Z_L = 0{,}5107 \cdot 50\Omega = 25{,}5\Omega \qquad \underline{\underline{L_s}} = \frac{X_L}{2\pi \cdot f} = \frac{25{,}5\,\frac{\mathrm{V}}{\mathrm{A}}}{2\pi \cdot 183 \cdot 10^6\,\frac{1}{\mathrm{s}}} = \underline{\underline{22{,}2\,\mathrm{nH}}}$$

$$B_C = \frac{0 - \left(-1{,}558\right)}{Z_L} = 31{,}6\,\mathrm{mS} \qquad \underline{\underline{C_p}} = \frac{B_{C_p}}{2\pi \cdot f} = \frac{31{,}6 \cdot 10^{-3}\,\frac{\mathrm{A}}{\mathrm{V}}}{2\pi \cdot 183 \cdot 10^6\,\frac{1}{\mathrm{s}}} = \underline{\underline{27{,}4\,\mathrm{pF}}}$$

**Aufgabe 2:** (25 Punkte)

1. ablesen aus Folie 3-6 mit $Z_L = 50\Omega$ und $\varepsilon_r = 4,0$: $\frac{w}{h} = 2,0$

$\underline{\underline{w}} = 2,0 \cdot h = 2,0 \cdot 0,8\text{mm} = \underline{\underline{1,6\text{mm}}}$

ablesen aus Folie 3-5 für $\frac{w}{h} = 2,0$: $\varepsilon_{r,eff} = 3,1$

$$\underline{\underline{\lambda}} = \frac{\lambda_0}{\sqrt{\varepsilon_{r,eff}}} = \frac{c_0}{f \cdot \sqrt{\varepsilon_{r,eff}}} = \frac{3 \cdot 10^8 \,\frac{\text{m}}{\text{s}}}{2,4 \cdot 10^9 \,\frac{1}{\text{s}} \cdot \sqrt{3,1}} = \underline{\underline{71,0\text{mm}}}$$

2. abgelesen aus Folie 7-14, da worst case = $\lambda/4$ - Koppelstrecke:

$C_{min} = 30\text{dB} \quad \rightarrow \quad \frac{s}{h} = 3,8 \quad \rightarrow \quad \underline{\underline{s_{min}}} = 3,8 \cdot h = 3,8 \cdot 0,8\text{mm} = \underline{\underline{3,04\text{mm}}}$

3. aus Folie 7-12 mit $\frac{s}{h} = \frac{3,2\text{mm}}{0,8\text{mm}} = 4,0$: $k_L = 0,03$

$$\left|\underline{s}_{41}\right|^2 = \frac{k_L^2 \cdot \sin^2 \beta l}{\left(1 - k_L^2\right) \cdot \cos^2 \beta l + \sin^2 \beta l} = \frac{k_L^2 \cdot \sin^2 \beta l}{-k_L^2 \cdot \cos^2 \beta l + \cos^2 \beta l + \sin^2 \beta l} =$$

$$= \frac{k_L^2 \cdot \sin^2 \beta l}{1 - k_L^2 \cdot \cos^2 \beta l} = \frac{(0,03)^2 \cdot \sin^2\left(\frac{2\pi}{71,0\text{mm}} \cdot 69,0\text{mm}\right)}{1 - (0,03)^2 \cdot \cos^2\left(\frac{2\pi}{71,0\text{mm}} \cdot 69,0\text{mm}\right)} = 2,792 \cdot 10^{-5} \qquad \left[7,238 \cdot 10^{-6}\right]$$

$$\underline{\underline{C}} = -10\text{dB} \cdot \lg\left(\left|\underline{s}_{41}\right|^2\right) = -10\text{dB} \cdot \lg\left(2,792 \cdot 10^{-5}\right) = \underline{\underline{45,5\text{dB}}} \qquad [51,4\text{dB}]$$

4. $C = 7,0\text{dB} \quad \rightarrow \; k = 10^{-\frac{7,0\text{dB}}{20\text{dB}}} = 0,4467$

$$Z_{L1} = Z_L \cdot \sqrt{1 - k^2} = 50\Omega \cdot \sqrt{1 - 0,4467^2} = 44,7\Omega$$

$$Z_{L2} = Z_L \cdot \frac{\sqrt{1 - k^2}}{k} = 50\Omega \cdot \frac{\sqrt{1 - 0,4467^2}}{0,4467} = 100\Omega$$

aus Folie 3-6:

$\frac{w_1}{h} = 2,5 \quad \rightarrow \; \underline{\underline{w_1}} = 2,5 \cdot 0,8\text{mm} = \underline{\underline{2,0\text{mm}}}$

$\frac{w_2}{h} = 0,49 \quad \rightarrow \; \underline{\underline{w_2}} = 0,49 \cdot 0,8\text{mm} = \underline{\underline{0,39\text{mm}}}$

**Aufgabe 3:** (20 Punkte)

1. $$\underline{\underline{v_{Tr}}} = 10\text{dB} \cdot \lg\left|\underline{s}_{21}\right|^2 = 10\text{dB} \cdot \lg\left(3{,}876^2\right) = \underline{\underline{11{,}8\text{dB}}}$$

$$\underline{\underline{v_{Tr,max}}} = 10\text{dB} \cdot \lg\left(\left|\underline{s}_{21}\right|^2 \cdot \frac{1}{1-\left|\underline{s}_{11}\right|^2} \cdot \frac{1}{1-\left|\underline{s}_{22}\right|^2}\right) = 10\text{dB} \cdot \lg\left(3{,}876^2 \cdot \frac{1}{1-0{,}614^2} \cdot \frac{1}{1-0{,}473^2}\right) = \underline{\underline{14{,}9\text{dB}}}$$

2. $$\underline{Z}_e = Z_L \cdot \frac{1+\underline{s}_{11}}{1-\underline{s}_{11}} = Z_L \cdot \frac{1+0{,}614 \cdot e^{-j152°}}{1-0{,}614 \cdot e^{-j152°}} = 17{,}2\Omega \cdot e^{-j43°} = \left(12{,}58 - j11{,}73\right)\Omega$$

$$\underline{\underline{\underline{Z}_{i,q}}} = \underline{Z}_e^* = \underline{\underline{17{,}2\Omega \cdot e^{+j43°}}} = \left(12{,}58 + j11{,}73\right)\Omega$$

$$\underline{Z}_a = Z_L \cdot \frac{1+\underline{s}_{22}}{1-\underline{s}_{22}} = Z_L \cdot \frac{1+0{,}473 \cdot e^{-j67°}}{1-0{,}473 \cdot e^{-j67°}} = 68{,}3\Omega \cdot e^{-j48°} = \left(45{,}70 - j50{,}76\right)\Omega$$

$$\underline{\underline{\underline{Z}_{Last}}} = \underline{Z}_a^* = \underline{\underline{68{,}3\Omega \cdot e^{+j48°}}} = \left(45{,}70 + j50{,}76\right)\Omega$$

3. $$B = 1 + \left|\underline{s}_{11}\right|^2 - \left|\underline{s}_{22}\right|^2 - \left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|^2 = 0{,}01$$

$$\left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right| = \sqrt{1-0{,}01+\left|\underline{s}_{11}\right|^2 - \left|\underline{s}_{22}\right|^2} = \sqrt{0{,}99+0{,}614^2-0{,}473^2} = 1{,}069$$

$$\left|0{,}614 \cdot e^{-j152°} \cdot 0{,}473 \cdot e^{-j67°} - \left|\underline{s}_{12}\right| \cdot e^{j20°} \cdot 3{,}876 \cdot e^{-j239°}\right| =$$

$$= \left|0{,}290 \cdot e^{-j219°} - 3{,}876 \cdot \left|\underline{s}_{12}\right| \cdot e^{-j219°}\right| = \left|0{,}290 - 3{,}876 \cdot \left|\underline{s}_{12}\right|\right| = 1{,}069$$

2 Lösungen:

a) $0{,}290 - 3{,}876 \cdot \left|\underline{s}_{12}\right| = 1{,}069 \qquad \left|\underline{s}_{12}\right| = \dfrac{-1{,}069+0{,}290}{3{,}876} = -0{,}201$ physikalisch sinnlos

b) $-0{,}290 + 3{,}876 \cdot \left|\underline{s}_{12}\right| = 1{,}069 \qquad \underline{\underline{\left|\underline{s}_{12}\right|}} = \dfrac{1{,}069+0{,}290}{3{,}876} = \underline{\underline{0{,}351}}$

4. $$K = \frac{1-\left|\underline{s}_{11}\right|^2 - \left|\underline{s}_{22}\right|^2 + \left|\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}\right|^2}{2\left|\underline{s}_{12}\underline{s}_{21}\right|} =$$

$$= \frac{1-0{,}614^2-0{,}473^2+\left|0{,}614 \cdot e^{-j152°} \cdot 0{,}473 \cdot e^{-j67°} - 0{,}351 \cdot e^{j20°} \cdot 3{,}876 \cdot e^{-j239°}\right|^2}{2 \cdot 0{,}351 \cdot 3{,}876} =$$

$= 0{,}568 < 1 \qquad \rightarrow$ nicht absolut stabil! $\qquad \left[0{,}610\right]$

**Aufgabe 4:** (19 Punkte)

1. $$P_{RA} = k \cdot T_A \cdot B = 1{,}38 \cdot 10^{-23}\,\tfrac{\text{Ws}}{\text{K}} \cdot 290\text{K} \cdot 100 \cdot 10^6\,\tfrac{1}{\text{s}} = 4{,}0 \cdot 10^{-13}\,\text{W} \mathrel{\hat{=}} -94\text{dBm}$$

$$\underline{\underline{\left(\frac{S}{N}\right)_A}} = \left(\frac{P_A}{P_{RA}}\right) \mathrel{\hat{=}} -80\text{dBm} - \left(-94\text{dBm}\right) = \underline{\underline{14\text{dB}}}$$

2. $$F_{VV} = 10^{\frac{f_{VV}}{10\text{dB}}} = 10^{\frac{1{,}8\text{dB}}{10\text{dB}}} = 1{,}51$$

$$F_L = 10^{\frac{a_L \cdot l}{10\text{dB}}} \qquad G_L = 10^{-\frac{a_L \cdot l}{10\text{dB}}} = \frac{1}{F_L}$$

$$F_{ges} = 1 + \left(F_L - 1\right) + \frac{F_{VV} - 1}{G_L} = \left(\frac{S}{N}\right)_A \Big/ \left(\frac{S}{N}\right)_{min} \mathrel{\hat{=}} 14\text{dB} - 10\text{dB} = 4\text{dB} \mathrel{\hat{=}} 2{,}51 \qquad \left[5\text{dB} \mathrel{\hat{=}} 3{,}16\right]$$

$$F_{ges} = F_L + \left(F_{VV} - 1\right) \cdot F_L = F_{VV} \cdot F_L$$

$$F_L = \frac{F_{ges}}{F_{VV}} = \frac{2{,}51}{1{,}51} = 1{,}66 \mathrel{\hat{=}} 2{,}2\text{dB} = a_L \cdot l \qquad \left[2{,}09 \mathrel{\hat{=}} 3{,}2\text{dB}\right]$$

$$\underline{\underline{l}} = \frac{2{,}2\text{dB}}{a_L} = \frac{2{,}2\text{dB}}{0{,}5\,\frac{\text{dB}}{\text{m}}} = \underline{\underline{4{,}4\text{m}}} \qquad \left[6{,}4\text{m}\right]$$

Kontrolle, ob das Signal dabei nicht unter den Minimalpegel gedämpft wird:

$$P_A - a_L + g_{VV} = -80\text{dBm} - 2{,}2\text{dB} + 20\text{dB} = -62{,}2\text{dBm} > -70\text{dBm} = P_{E,min} \qquad \left[-63{,}2\text{dBm}\right]$$

3. $$F_L = 10^{\frac{0{,}5\frac{\text{dB}}{\text{m}} \cdot 20\text{m}}{10\text{dB}}} = 10 \qquad G_L = 10^{-\frac{0{,}5\frac{\text{dB}}{\text{m}} \cdot 20\text{m}}{10\text{dB}}} = 0{,}1 \qquad G_{VV2} = 10^{\frac{10\text{dB}}{10\text{dB}}} = 10 \qquad F_{ges} = 2{,}5$$

$$F_{ges} = 1 + \left(F_{VV2} - 1\right) + \frac{F_L - 1}{G_{VV2}} + \frac{F_{VV} - 1}{G_{VV2} \cdot G_L}$$

$$F_{VV2} = F_{ges} - \frac{F_L - 1}{G_{VV2}} - \frac{F_{VV} - 1}{G_{VV2} \cdot G_L} = 2{,}5 - \frac{10 - 1}{10} - \frac{1{,}51 - 1}{10 \cdot 0{,}1} = 2{,}5 - 0{,}9 - 0{,}51 = 1{,}09$$

$$\rightarrow \quad \underline{\underline{f_{VV2} = 0{,}37\text{dB}}}$$

**Aufgabe 5:** (16 Punkte)

1.

$$\underline{b}_{Q2} = \underline{b}_Q \cdot \frac{\underline{s}_{21}}{1-\underline{s}_{11}\cdot\underline{r}_Q} = \underline{b}_Q \cdot \frac{0,30}{1-\left(-j0,70\right)\cdot j0,80} = \underline{b}_Q \cdot 0,6818$$

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12}\cdot\underline{s}_{21}\cdot\underline{r}_Q}{1-\underline{s}_{11}\cdot\underline{r}_Q} = -j0,70 + \frac{0,30\cdot 0,30\cdot j0,80}{1-\left(-j0,70\right)\cdot j0,80} = -j0,5364$$

$$\underline{\underline{P_L}} = \tfrac{1}{2}\left|\underline{b}_{Q2}\right|^2 \cdot \frac{1-\left|\underline{r}_L\right|^2}{\left|1-\underline{r}_2\cdot\underline{r}_L\right|^2} = \tfrac{1}{2}\left|\underline{b}_Q\right|^2 \cdot 0,6818^2 \cdot \frac{1-0,50^2}{\left|1-\left(-j0,5364\right)\cdot j0,50\right|^2} = 4,649\text{W}\cdot 1,400 = \underline{\underline{6,51\text{W}}}$$

2.

$$P_{L,abs,0} = \tfrac{1}{2}\left|\underline{b}_Q\right|^2 \cdot \frac{1-\left|\underline{r}_L\right|^2}{\left|1-\underline{r}_Q\cdot\underline{r}_L\right|^2} = 10\text{W}\cdot\frac{1-0,50^2}{\left|1-j0,80\cdot j0,50\right|^2} = 3,83\text{W}$$

$$\underline{\underline{a_E}} = 10\text{dB}\cdot\lg\left(\frac{P_{L,abs,0}}{P_L}\right) = 10\text{dB}\cdot\lg\left(\frac{3,83\text{W}}{6,51\text{W}}\right) = \underline{\underline{-2,3\text{dB}}} \qquad \left[-1,9\text{dB}\right]$$

3.

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12}\cdot\underline{s}_{21}\cdot\underline{r}_L}{1-\underline{s}_{22}\cdot\underline{r}_L} = -j0,70 + \frac{0,30\cdot 0,30\cdot j0,50}{1-\left(-j0,7\right)\cdot j0,50} = -j0,6308$$

$$P_{Q,ab} = \tfrac{1}{2}\left|\underline{b}_Q\right|^2 \cdot \frac{1-\left|\underline{r}_1\right|^2}{\left|1-\underline{r}_Q\cdot\underline{r}_1\right|^2} = 10\text{W}\cdot\frac{1-0,6308^2}{\left|1-j0,80\cdot\left(-j0,6308\right)\right|^2} = 24,5\text{W}$$

$$\underline{\underline{a_B}} = 10\text{dB}\cdot\lg\left(\frac{P_{Q,ab}}{P_L}\right) = 10\text{dB}\cdot\lg\left(\frac{24,5\text{W}}{6,51\text{W}}\right) = \underline{\underline{5,8\text{dB}}} \qquad \left[6,1\text{dB}\right]$$